

SARDAR PATEL UNIVERSITY

B.Sc. (CA&IT) (NC) / M.Sc IT Integrated (NC) Examination

2nd Semester (CBCS)

Tuesday , Date: 2/4/2019

Session : Morning Time : 10:00 am to 1:00 pm

Subject Code: PS02EHT01

Subject Title : Digital Electronics

Total Marks: 70

Q1. Multiple Choice Questions.

[10]

1. A combinational circuit that performs the arithmetic addition of two bits is called _____.

a. Full Adder	b. Half Adder
c. Decoder	d. Binary Adder
2. A _____ is a combinational circuit that converts binary information from the 2^n coded inputs to an output.

a. Full Adder	b. Half Adder
c. Encoder	d. Decoder
3. In half adder XOR gate's output is _____.

a. remainder	b. carry
c. sum	d. None of the above
4. Which is used as data selector?

a. Encoder	b. Decoder
c. Half Adder	d. Subtractor
5. Which device has many input and one outputs?

a. Multiplexer	b. Demultiplexer
c. Counter	d. Flip Flop
6. A Comparator compare how many word?

a. One	b. Two
c. Three	d. Four
7. In k-map, pair eliminates _____ variable.

a. One	b. Two
c. Three	d. Four
8. The Boolean expression $A + B' + C$ is _____ term.

a. sum	b. Linear
c. product	d. Complemented
9. Which of the following is known as Universal flip-flop ?

a. RS flip-flop	b. Master JK flip-flop
c. JK flip-flop	d. D flip-flop
10. In shift left register, the arrival of the first rising clock edge sets the _____ flip-flop.

a. Up	b. Down
c. Left	d. Right

(P.T.O.)

Q2. Answer the following short questions (Attempt any TEN) [20]

1. Explain full adder in brief.
2. Explain binary adder in brief.
3. Explain binary subtractor in brief.
4. Draw the circuit of Nibble multiplexer.
5. Draw the circuit of Comparator.
6. Draw the circuit of 1x4 line de-multiplexer
7. Describe pair in k-map in brief.
8. Explain Sum of Product in detail.
9. Define Karnaugh map .
10. Draw circuit diagram of shift left register.
11. Define flip-flop
12. Draw circuit diagram of ring counter.

Q3.a. Explain 8x3 line encoder in detail. [05]
b. Explain De-Morgan's first and second theorem. [05]

OR

Q3.a. Explain 3x8 line decoder in detail. [05]
b. Explain half adder in detail. [05]

Q4.a. Write a short note on 7 Segment decoder. [05]
b. Explain 4x1 line multiplexer with circuit in detail. [05]

OR

Q4.a. Write a short note on Comparator with circuit diagram. [05]
b. Explain 1x8 line de-multiplexer with circuit in detail. [05]

Q5.a. Simplify this using k-map $F(A,B,C,D)=\sum(1,3,5,6,8,11,15)$ [05]
b. What is k-map? Explain pair and quad with example. [05]

OR

Q5.a. Simplify this using k-map $F(A,B,C,D)=\sum(1,2,5,6,8,12,14)$ [05]
b. Write a short note on Don't Care Condition. [05]

Q6. Explain following in detail with necessary diagrams: [10]
i) D flip-flop ii) RS Flip-flop

OR

Q6. Explain following in detail with necessary diagrams: [10]
i) Shift Right register ii) Controlled buffer register

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